WORKED SOLUTION

STEP II 1997, Question 8

Pure • Inequalities by Integration

STEP II 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q8

The key idea

Each expression of the form $(x^p - 1)/p$ is an integral in disguise: it is $\int_1^x t^{p-1}\,\mathrm{d}t$. So comparing two such expressions comes down to comparing the integrands t^{p-1} and t^{q-1}, which is easy once we notice that the comparison flips at $t = 1$. The final inequality is then one more integration, this time of the inequality we have just proved.

Solution

Integrating an inequality

If $\mathrm{f}(t) \geq \mathrm{g}(t)$ on $[a, b]$, then $\mathrm{f} - \mathrm{g} \geq 0$ there, so the region between the graph of $\mathrm{f} - \mathrm{g}$ and the t-axis lies on or above the axis and its signed area is non-negative:

$$\int_a^b \mathrm{f}(t)\,\mathrm{d}t - \int_a^b \mathrm{g}(t)\,\mathrm{d}t = \int_a^b (\mathrm{f}(t) - \mathrm{g}(t))\,\mathrm{d}t \geq 0.$$

Note that this needs $a \leq b$: if the limits are the wrong way round, the inequality between the integrals reverses.

The case $x \geq 1$

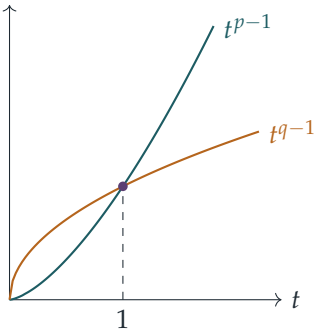
For $t \geq 1$ and $p > q$ we have $t^{p-1} = t^{q-1}\,t^{p-q} \geq t^{q-1}$, because $t^{p-q} \geq 1$ when $t \geq 1$ and $p - q > 0$. Since $x \geq 1$, the interval $[1, x]$ has its limits in the right order, so by the first part

$$\int_1^x t^{p-1}\,\mathrm{d}t \;\geq\; \int_1^x t^{q-1}\,\mathrm{d}t.$$

As $p, q > 0$, both integrals can be evaluated directly:

$$\int_1^x t^{p-1}\,\mathrm{d}t = \left[\frac{t^p}{p}\right]_1^x = \frac{x^p - 1}{p}, \qquad \int_1^x t^{q-1}\,\mathrm{d}t = \frac{x^q - 1}{q}.$$

Hence $\dfrac{x^p - 1}{p} \geq \dfrac{x^q - 1}{q}$ for $x \geq 1$, as required.



The sketch (drawn with $p = 2.5$, $q = 1.5$) shows the point: the larger power wins for $t > 1$ and loses for $0 < t < 1$.

The case $0 \le x \le 1$

Ask yourself For $0 < t < 1$ the inequality between t^{p-1} and t^{q-1} goes the other way. Why does the conclusion not reverse too?

Two reversals cancel. For $0 < t \le 1$ we have $t^{p-q} \le 1$, so $t^{p-1} \le t^{q-1}$. But now $x \le 1$, so the interval in the right order is $[x, 1]$, not $[1, x]$.

First take $0 < x \le 1$. Integrating $t^{q-1} \ge t^{p-1}$ over $[x, 1]$ (both are continuous there, as $x > 0$) gives

$$\int_x^1 t^{q-1}\,\mathrm{d}t \;\ge\; \int_x^1 t^{p-1}\,\mathrm{d}t, \qquad \text{that is,} \qquad \frac{1-x^q}{q} \;\ge\; \frac{1-x^p}{p}.$$

Multiplying by -1 reverses the inequality, giving $\dfrac{x^p-1}{p} \ge \dfrac{x^q-1}{q}$.

For $x = 0$ we check directly: the inequality reads $-\frac{1}{p} \ge -\frac{1}{q}$, which is true because $p > q > 0$ gives $\frac{1}{p} < \frac{1}{q}$. (We treat $x = 0$ separately because if $q < 1$ then t^{q-1} is unbounded near 0.) So the inequality holds for all $0 \le x \le 1$, and together with the previous case, for every $x \ge 0$.

The final inequality

Write $\phi_p(t) = \dfrac{t^p - 1}{p}$. We have now shown that $\phi_p(t) \ge \phi_q(t)$ for every $t \ge 0$ whenever $p > q > 0$. Both functions are continuous on $[0, \infty)$, so for $x > 0$ we may integrate this inequality over $[0, x]$:

$$\int_0^x \frac{t^p-1}{p}\,\mathrm{d}t \;\ge\; \int_0^x \frac{t^q-1}{q}\,\mathrm{d}t.$$

Now

$$\int_0^x \frac{t^p-1}{p}\,\mathrm{d}t = \frac{1}{p}\Big(\frac{x^{p+1}}{p+1} - x\Big) = \frac{x}{p}\Big(\frac{x^p}{p+1} - 1\Big),$$

and similarly for q. So

$$\frac{x}{p}\Big(\frac{x^p}{p+1}-1\Big) \;\geq\; \frac{x}{q}\Big(\frac{x^q}{q+1}-1\Big),$$

and dividing by $x > 0$ (which keeps the direction of the inequality) gives

$$\frac{1}{p}\Big(\frac{x^p}{p+1}-1\Big) \;\geq\; \frac{1}{q}\Big(\frac{x^q}{q+1}-1\Big).$$

At $x = 0$ this reads $-\frac{1}{p} \geq -\frac{1}{q}$, which we have already seen is true. This completes the proof for all $x \geq 0$.

Checks

- At $x = 1$ the middle inequality is $0 \geq 0$: the two curves cross there, consistent with equality.
- Take $p = 2$, $q = 1$, $x = 3$ in the final inequality: the left side is $\frac{1}{2}(3-1) = 1$ and the right side is $\frac{3}{2} - 1 = \frac{1}{2}$, so $1 \geq \frac{1}{2}$.
- A random test of 200,000 triples (p, q, x) with $0 < q < p$ and $x \geq 0$, including $x = 0$, $x = 1$ and tiny x, found no counterexample to either inequality.

Common slips

- Integrating over $[1, x]$ when $x < 1$ and keeping the inequality the same way round: the limits must be in increasing order before you integrate an inequality.
- Forgetting the end point $x = 0$, where t^{q-1} may not even be defined.
- Trying to prove the last part from scratch by calculus, rather than integrating the inequality you have just established.

Beyond STEP

As $q \to 0$, $(x^q - 1)/q \to \ln x$, so the result shows that $(x^p - 1)/p$ decreases to $\ln x$ as p decreases to 0: a neat family of bounds on the logarithm, which is also the basis of the Box–Cox transformation in statistics. The principle that integration preserves inequalities (monotonicity of the integral) is one of the first theorems proved about the Riemann integral in an analysis course. See Abbott, *Understanding Analysis*.

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